Adaptive Feature Fusion With Attention-Guided Small Target Detection in Remote Sensing Images

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***Abstract*— Small target detection in remote sensing images has considerable significance in practical applications such as military dynamic discrimination and traffic monitoring. However, the limited appearance features of small-scale targets and the widespread false alarm sources make small target detection in remote sensing images a tough challenge. To address these problems, we propose a novel small detection method by employing an adaptive multilevel feature fusion module (AMFFM) and an attention-augmented high-resolution head (AAHRH). Specifically, AMFFM is designed to suppress the interference of false alarm sources in complicated scenes. We upsample the high-level features by the context modeling of semantic information and refine the low-level features for noise removal. Then the enhanced multilevel features are fused based on the spatial and channel significance. After that, AAHRH is put forward to enhance the perception of small targets by embedding cross-dimension interaction with the attention mechanism. The prediction heads are reconstructed with high-resolution layers to improve the detection performance in densely distributed scenes. We conduct dilated and comparison experiments on a constructed small car dataset, a public small ship dataset, and the VEDAI dataset. The experimental results on two datasets verify the effectiveness and robustness of the proposed method with the state-of-the-art performance.**

***Index Terms*— Adaptive feature fusion, cross-dimension attention mechanism, optical remote sensing, small target detection.**

I. Introduction

REAL-TIME and accurate detection of small-scale targets like cars and ships in remote sensing images is of great significance for military and civilian applications such as key area surveillance, military dynamic discrimination, traffic management, and planning [1], [2], [3], [4]. Currently, optical remote sensing images have emerged as the primary input data for detection models owing to their extensive detection range, diverse spectral characteristics, and high image resolution [5]. The development of optical remote sensing techniques makes detecting or even recognizing small targets possible.

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However, limited by the imaging resolution and quality, targets such as cars and ships exhibit the characteristics of small geometric scales and few appearance features. The lack of detailed information, such as geometric texture and edge contours, makes it difficult to characterize target features [6]. At the same time, various false alarm sources exist whose shape and texture are similar to those of small targets in complex scenes [7], such as waves and reefs for ships, shadows, and roofs for vehicles. The above challenges make it tough for existing methods to meet the detection requirements with low false alarm rates and high detection accuracy. Therefore, it is urgent to conduct research on fast and accurate small target detection methods combined with practical requirements in remote sensing applications [8].

Traditional detection methods depend on the hand-crafted feature in the shallow levels, which are difficult to achieve effective feature characterization of small targets in complicated scenes. With the development of deep learning technology, convolutional neural networks (CNNs) are gradually used to detect in remote sensing scenes. Through trainable feature extractors, deep learning methods can mine semantic features and perform better than traditional methods [9]. Among them, the two-stage networks generate candidate proposals, classify the generated regions, and regress the target coordinates. Deng et al. [10] improve the faster R-CNN with the multiscale object proposal network (MS-OPN) and accurate object detection network (AODN) to achieve an effective performance for large-scale remote sensing images. Shivappriya et al. [11] apply the additive activation function in the faster R-CNN, which has the advantage of better convergence and clear bounding variance. The one-stage networks directly output the target category probabilities and location coordinates without region proposal. They have the potential to achieve real-time detection compared to the two-stage networks. Lu et al. [12] combine a multilayer feature fusion structure to enhance the semantic information and a dual-path attention module to screen the feature with SSD, which achieves effective performance in public remote sensing datasets. Wang et al. [13] design a saliency-guided dense sampling layer (SDSL) and a saliency region-aware convolution (SAConv) strategy to improve the representation capability for YOLO. The existing methods mainly introduce feature fusion and attention mechanisms to achieve a better response for small targets. Feature fusion considers both shallow appearance and deep context features simultaneously, which helps to suppress the interference of false alarms source.

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Furthermore, attention mechanisms adjust the importance of multiple features, thus improving the perception ability of targets. However, existing feature fusion methods integrate the multilayer features directly, which ignores that the detailed low-level and semantic high-level features contribute differently to the small target detection. Besides, the attention mechanisms only converge the information of a single dimension when generating the attention weights, which cannot capture the long-range dependencies between different dimensions.

Therefore, we propose a novel small target detection method based on CNNs to tackle with the difficulties caused by the limited appearance of small targets and widespread false alarm sources. Our approach is combined with the adaptive feature fusion and cross-dimension attention mechanism to achieve effective detection performance while ensuring high inference speed. First, we construct an adaptive multilevel feature fusion module (AMFFM) to weaken the interference of false alarm sources in complex scenes. The module dynamically adjusts the upsampling weight kernel to achieve semantic modeling of high-level features. At the same time, the low-level feature is refined to remove the noise disturbance and fused with the high-level features by adaptive fusion weights. Then, we put forward an attention-augmented high-resolution head (AAHRH) to fully exploit the valid features. It helps to enhance the perceptual capability of small targets and improve detection performance in densely distributed scenes. The application performance is verified by a constructed small car dataset, a public small ship dataset, and the VEDAI dataset. The experimental results prove the effectiveness and robustness of the proposed method, which can achieve optimal detection performance and ensure a high reference speed.

The main contributions of this work can be summarized as follows.

1) A novel small target detection method combined with the adaptive feature fusion and cross-dimension attention mechanism is proposed, which can meet the requirements of high detection accuracy with promising inference speed.
2) An AMFFM is constructed to suppress the interference of false alarm sources whose characteristics are similar to those of targets in complicated scenes, which is achieved by the context modeling of semantic features and the noise removal of detailed features combined with the generation of adaptive fusion weights.
3) An AAHRH is put forward to enhance the perception of small targets by embedding cross-dimension interaction with the attention mechanism, and the prediction level is reconstructed with the high-resolution layer to perform better in densely distributed scenes.

The remainder of this article is organized as follows. Section II illustrates the related works and existing problems of small target detection. In Section III, we elaborate on the rationale and details of the proposed architecture. We conduct the dilated and comparison experiments to verify the effectiveness of the proposed method in Section IV. Finally, we summarize and conclude in Section V.

II. Related Works

In this section, we first briefly introduce the development of small target detection technology and point out the shortcoming of current methods. We then review the feature fusion method and attention mechanism. On this basis, we explain the idea of our proposed method.

A. Small Target Detection Method

Small target detection in remote sensing images often faces two challenges, one of which is that the small scale and weak characteristics of targets undermine the feature representation of the network. Usually, the target occupies a small proportion of an image, and the limited appearance information makes it difficult for the network to extract discriminative features. The other one is that there are various false alarm sources with similar features to small targets in the remote sensing images. The above difficulties make small target detection in remote sensing images far more challenging than generic target detection. The direct use of the existing detection paradigms cannot obtain promising performance, which means strategies and modules must be designed for the specific application.

According to whether generating the candidate regions, target detection networks can be broadly divided into one-stage and two-stage detectors. Two-stage detectors usually perform better for generic target detection by virtue of the coarse-to-fine framework. Ji et al. [14] modify the faster R-CNN with the bottom-up feature extraction and top-down multiscale feature fusion for the region proposal generation with small vehicles. Luo et al. [15] apply neural architecture search (NAS) to obtain the optimal cross-layer connection based on faster R-CNN while using the multilayer information to enrich the feature of vehicles. Compared to two-stage detectors, one-stage detectors directly regress the probability and coordinates of the target with an obvious inference speed advantage. SSD [16], the first method considering the improvement of small target detection, utilizes multilevel feature maps to predict where the shallow layer is used to detect small targets. DSSD [17] adds a deconvolution layer to recover image details and enhance the discrimination of small targets while suppressing irrelevant interference. FA-SSD [18] incorporates a multiscale feature fusion method in SSD to exploit contextual information. Besides, it integrates the attention mechanism in the shallow feature layer to enhance the ability to perceive small targets. Based on RetinaNet [19], QueryDet [20] takes the computational redundancy of low-level features and feature pyramid into consideration and then uses the cascade query strategy to achieve repaid small target detection. For YOLO-based architecture, the additional high-resolution prediction layer allows YOLOv3 [21] to obtain great performance for small target detection. YOLO-ACN [22] is constructed based on YOLOv3 by adding the channel and spatial attention mechanism in the residual blocks to focus on small targets and unitizing Soft-NMS to filter out the more accurate bounding boxes in dense-occluded scenes. Zhou et al. [3] propose a feature fusion structure IR-PANet to improve the recognition capability for small targets and apply the gamma correction to brighten the shadow part of the image. Inspired by the existing

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methods, the proposed approach combines the feature fusion and attention mechanism with the YOLO-based framework to ensure the balance of speed and accuracy for small target detection in remote sensing images.

B. Feature Fusion Method

CNN generates hierarchical feature maps with multiple spatial resolutions [8]. The shallow feature maps contain detailed features and localization information of the small targets. While the deep feature with semantic features help the network to perceive the comprehensive information of the images. Therefore, both shallow appearance and deep context features must be exploited simultaneously to achieve promising performance. Feature fusion is the solution to obtain a better response for small targets by integrating multilayer features. Feature pyramid network (FPN) [23] fuses the feature representations of different dimensions and passes the high-level information to the low-level layer through a top-down path. PANet [24] adds the bottom-up path on top of FPN to compensate for the lack of detailed information in the transmission process. The bidirectional paths consider both the semantic and localization features. NAS-FPN [25] applies NAS techniques to automatically get the optimal architecture based on reinforcement learning. Recursive-FPN [26] introduces additional feedback connections from the FPN to the backbone extraction network. Info-FPN [27] concentrates on the elimination solution of information loss from channel reduction, feature misalignment, and aliasing effect, thus proposing the pixel shuffle-based lateral connection module (PSM), feature alignment module (FAM), and semantic encoder module (SEM). Different from the existing approaches concerning the adjustment of fusion branches, the proposed method focuses on the adaptive fusion of multilevel features by upsampling with semantic modeling and refining the low-level features.

C. Attention Mechanism

Visual attention mechanism suppresses the negligible stimuli and quickly focuses on the significant regions in the scene, which is an important component of the human visual system. It helps locate the target of interest in complex backgrounds. Inspired by that, the attention mechanism is proposed to adjust the importance of multiple features by applying weights, which helps the network to enhance the perception ability of targets while suppressing irrelevant interferences. The attention mechanism can be roughly divided into the channel and spatial attention mechanism by the adjusted dimension. The channel attention mechanism automatically learns the significance of each channel and strengthens the semantic information. At the same time, the spatial attention mechanism generates a mask along the spatial dimension and assigns weights to enhance the perception of localization features. As a classical channel attention mechanism, squeeze-and-excitation (SE) [28] aggregates the spatial information globally in each channel through average pooling and uses fully connected layers to capture the relationships between channels. It generates the channel mask and helps to enhance the expression of specific semantic information. Convolutional block attention module (CBAM) [29], [30] combines a channel attention module (CAM) and a spatial attention module (SAM) in serial. The SAM integrates the channel information at each spatial location through average and maximum pooling, then uses convolution to generate the spatial significance. Inspired by the non-local operation, DANet [31] parallelizes the CAM and SAM, which pays attention to mining the global semantic information. Coordinate attention (CA) [32] embeds the positional information into the attention mechanism with little additional computation, which helps the networks more accurately locate the targets. Most attention-based modules for object detection in remote sensing are constructed from the attention mechanisms in the natural image process. Spatial groupwise enhancement (SGE) [33] is derived from SE but focuses on the partial channel information. It generates the attention factor for each spatial position in each semantic group to adjust the importance of features. Enhanced effective channel attention (ECCA) [34] is inspired by ECA, which utilizes two 1-D convolutional layers to capture nonlinear cross-channel interaction. Normalization-based attention module (NAM) [35] is the variant of CBAM, which reweight attention by adjusting the variance measurement of the training weight. Unlike the above approaches, the attention mechanism used in this article focuses on capturing the long-range dependencies between different dimensions to obtain a wide range of perceptual fields.

III. Proposed Method

A. Method Overview

The main architecture of the proposed method is illustrated in Fig. 1, which consists of a backbone feature extraction network, AMFFM, and AAHRH. The backbone feature extraction network adopts the same structure as YOLOv5 [36]. Considering that for small targets, excessive stacking of convolutional blocks only increases additional operations with little detection performance improvement. Therefore, we choose the backbone of YOLOv5n to reduce the computational complexity and fully maintain the speed advantage of one-stage network. The feature maps are downsampled from the images by 2×, 4×, 8×, 16×, and 32× (P1–P5). Then AMFFM is followed for adaptive refinement and fusion of the multilevel features, as described in Section III-B. After the fusion, AAHRH is introduced to implement the feature enhancement by attention mechanism and achieve the prediction by high-resolution P2 and P3 heads in Section III-C.

B. Adaptive Multilevel Feature Fusion Module

The low-level feature extracted by the backbone network contains detailed information of small targets, while the high-level is full of the semantic information of complex scenes. By fusing these multilevel features, the interference of false alarm sources with the targets can be effectively eliminated. The mainstream fusion methods can usually be decomposed into three steps. The first step comes to upsample the high-level features for spatial dimension expansion, followed by the lateral connection to introduce the low-level features. Finally, concatenating or addition is applied to fuse the multilevel features. However, the typical fusion methods such as FPN and PAN have problems in implementing the above steps.

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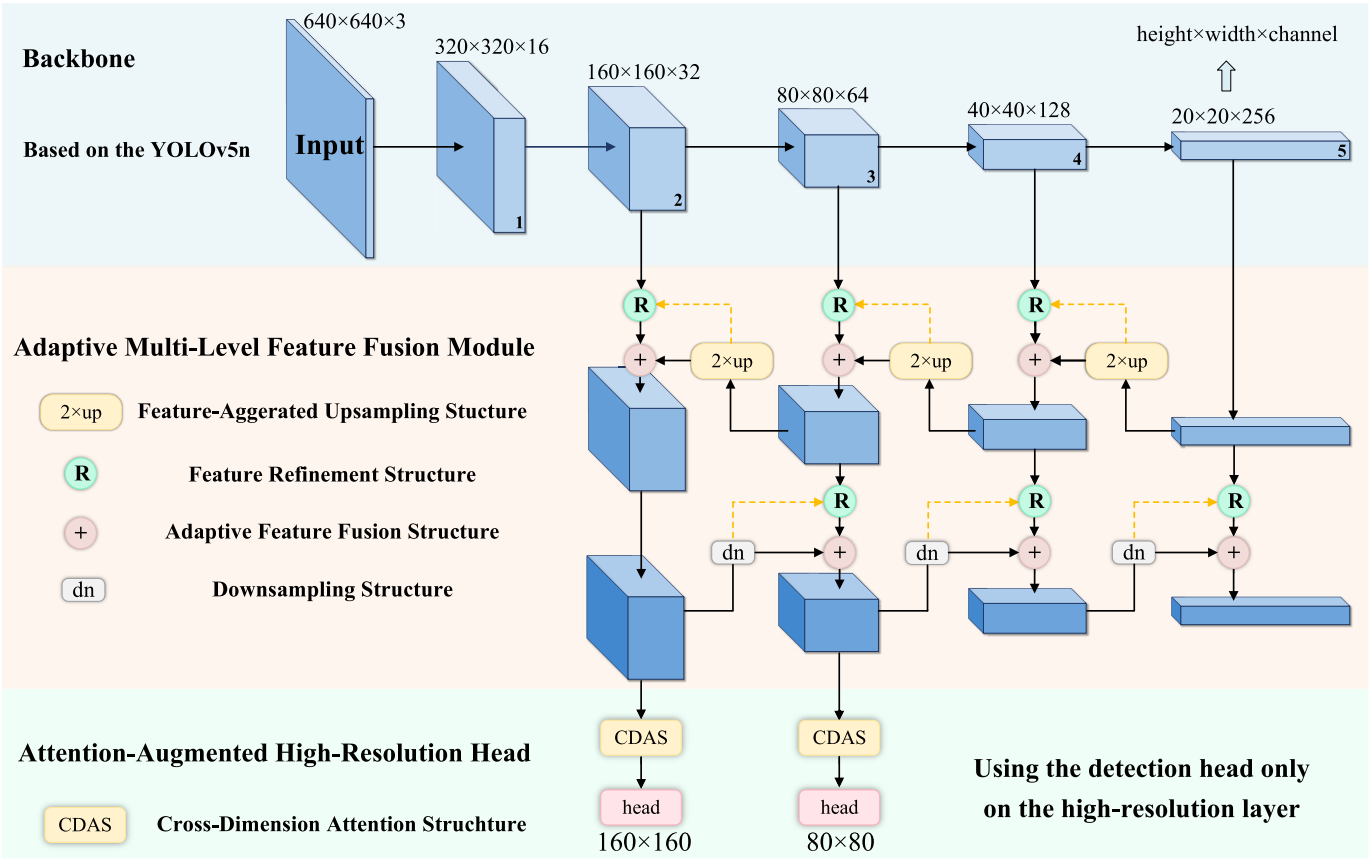


Fig. 1. Architecture of the proposed method. After extracting the multilevel features from the backbone, AMFFM is constructed to fuse the refined low-level features with the aggerated high-level features. Then, AAHRH embeds the cross-dimension attention with reconstructed detection heads to improve the perception of small targets.

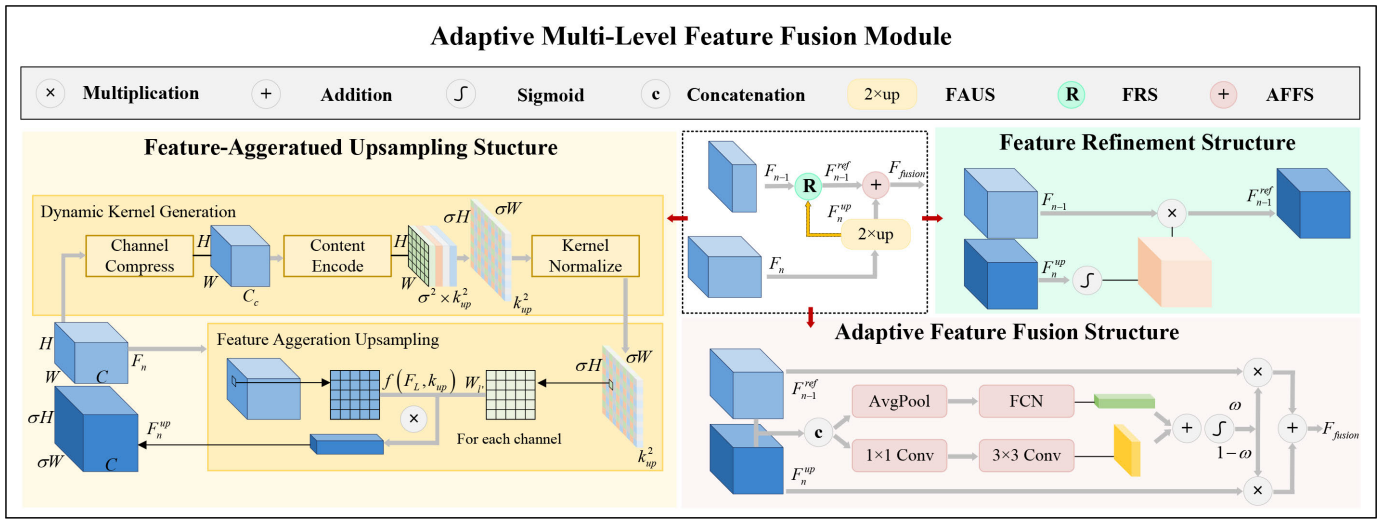


Fig. 2. Diagram of AMFFM. First, FAUS generates upsampled high-level features with dynamic kernels. Then, FRS refines low-level features with high-level features. Finally, features are fused adaptively through spatial and channel saliency.

1) The upsampling steps cannot make full use of the contextual information. The interpolation method has a limited receptive field, thus lacking global information. The deconvolution method learns the fixed weights of upsampling kernels, which are not spatially relevant and cannot utilize the semantic features.
2) Low-level features are often noisy due to insufficient extraction, interfering with the final feature fusion.
3) The fusion process achieved by addition is improper due to ignoring the importance of each level. Therefore, we propose an AMFFM, consisting of a feature-aggerated upsampling structure (FAUS), a feature refinement structure (FRS), and an adaptive feature fusion structure (AFFS) to solve the above problems. The diagram of AMFFM is shown in Fig. 2.

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The overall architecture of AMFFM adopts a similar fusion direction as PAN [24], which delivers high-level semantic information and low-level detailed information through top-down and bottom-up paths, respectively. First, the high-level features need to be upsampled to match the spatial dimension of the low-level features. To utilize the global information and generate more informative feature maps, we adopt the FAUS inspired by CARAFE [37]. This structure comprises two parts: dynamic kernel generation and feature aggregation upsampling. The former generates the upsampling kernels with different weights for each region based on the semantic information of the scene. Assume that the length, width, and channel dimension of the input feature map F_n are denoted by (H, W, C). We use convolution layers to compress the channels from C to C_c. The channel compressor helps to reduce the computational complexity of the subsequent processing. For the compressed features F_n^c, we exploit the contextual information within a larger receptive field through a convolutional layer with the kernel size of $k_{\text{encode}} \times k_{\text{encode}}$. The dimension of the encoded feature F_n^e becomes $(H, W, \sigma^2 \times k_{\text{up}}^2)$, where σ is the upsampling factor and k_{up} is the size of the neighborhood used for upsampling. To get the final dynamic kernel weights W, we reorganize the pixel of F_n^e and normalize it along the channel dimension, adjusting the dimensionality to $(\sigma H, \sigma W, k_{\text{up}}^2)$. The vectors along the channel dimension of W represent the generated upsampling kernel in each pixel. The kernels at different positions are dynamically adjusted based on contextual information to match spatial variability. After obtaining the dynamic kernel weights, we upsample the feature map by feature aggregation. The upsampled F_n^{up} in the kth channel and position (i', j') can be calculated

$$F_n^{\text{up}}(i', j', k) = \sum_{m=-\lfloor k_{\text{up}}/2 \rfloor + 1}^{\lfloor k_{\text{up}}/2 \rfloor} \sum_{n=-\lfloor k_{\text{up}}/2 \rfloor + 1}^{\lfloor k_{\text{up}}/2 \rfloor} W_{i'j'}(m, n) \times F_n(i + m, j + n, k) \tag{1}$$

where $F_n(i, j, k)$ is the value of the original feature map in the kth channel and position (i, j). The mapping relationship between (i, j) and (i', j') is $i = [i'/\sigma]$, $j = [j'/\sigma]$. $[\cdot]$ is the integer sign. $W_{i'j'}$ is the sampling kernel weight at the location (i', j'), which is obtained by extracting the channel vector of W and reshaping the dimension to $(k_{\text{up}}, k_{\text{up}})$. The final upsampled feature map F_n^{up} can be obtained by calculating for each position and each channel.

After upsampling high-level feature maps, the low-level feature maps need to be refined before fusing. Since the shallow feature is not sufficiently extracted by the CNN, there exists plenty of noise that affects the fusion for effective features. The high-level feature is rich in semantic information with little interference, so we propose the FRS to refine low-level features with high-level features. The reference map is obtained by processing the upsampled high-level feature F_n^{up} with the nonlinear activation function $\sigma(\cdot)$. Then it is applied to the low-level features F_{n-1} to correct the outliers. The modified feature map F_{n-1}^{ref} can be computed as

$$F_{n-1}^{\text{ref}} = F_{n-1} \times \sigma\left(F_n^{\text{up}}\right) \times F_n(i + m, j + n, k) \tag{2}$$

where $\sigma(\cdot)$ represents the sigmoid activation function.

The final step is to fuse the upsampled high-level features with the refined low-level features. Since the feature maps from multiple levels weigh differently for the feature characterization of small targets, the features rich in semantic information of scenes may interfere with the features of small targets if fused directly. Based on the feature saliency idea, we propose the AFFS. By assigning corresponding weights to the features of each level, we adjust their importance in the training and inference process. The structure generates fusion weights adaptively through spatial and channel saliency. First, we concatenate the low-level feature F_{n-1}^{ref} and the high-level feature F_n^{up}. The concatenated features F_n^C is used to generate the saliency weights. To get the channel saliency, we use spatial pooling to aggerate the global information. Then a fully connected network is followed to learn the dependencies between channels. Meanwhile, the channel number of F_n^C is adjusted by 1×1 convolution, and the local dependency is exploited by 3×3 convolution to generate spatial saliency. Finally, they are combined through the broadcast mechanism to generate the pixel-by-pixel feature saliency. Then the sigmoid activation function is followed to obtain the fusion weights. The weights generation can be formulated as

$$\omega = \sigma\left(f_{\text{FCN}}\left(\text{AvePool}\left(F_n^C\right)\right) + \text{conv}_{3\times3}\left(\text{conv}_{1\times1}\left(F_n^C\right)\right)\right) \tag{3}$$

where $\text{AvePool}(\cdot)$ is the average global pooling, $f_{\text{FCN}}(\cdot)$ is the fully connected network, $\text{conv}_{3\times3}$ and $\text{conv}_{1\times1}$ represent the 3×3 convolution and 1×1 convolution, respectively.

The final fusion process is adjusted by the weights ω adaptively, which is written as

$$F_{\text{fusion}} = \omega \times F_{n-1}^{\text{ref}} + (1 - \omega) \times F_n^{\text{up}}. \tag{4}$$

C. Attention-Augmented High-Resolution Head

The overall structure of AAHRH is illustrated in Fig. 3. Considering the limited geometric scale and weak characteristics of small targets in remote sensing images, we apply the attention mechanism to enhance the perception ability of the network before prediction. Also, the prediction layer with high resolution is utilized to improve the detection performance for densely distributed scenes.

The attention mechanism simulates the observation process of human visual systems, which focus on the saliency area in complicated scenes. The essence is to rescale the distribution of the input feature map based on the significance weights learned from the feature map itself. By assigning weights to different channels and regions, the network is prone to perceive specific detailed features and semantic information. However, the conventional attention mechanisms only converge the information of a single dimension when generating the attention weights, which cannot capture the long-range dependencies between different dimensions. The loss of cross-dimension interaction makes the network difficult to distinguish the small targets from the complicated false alarm source. Therefore, we propose the cross-dimension attention structure (CDAS), which concentrates on mining the dependencies of different dimensions and can obtain a wider receptive field. The receptive field represents the range of

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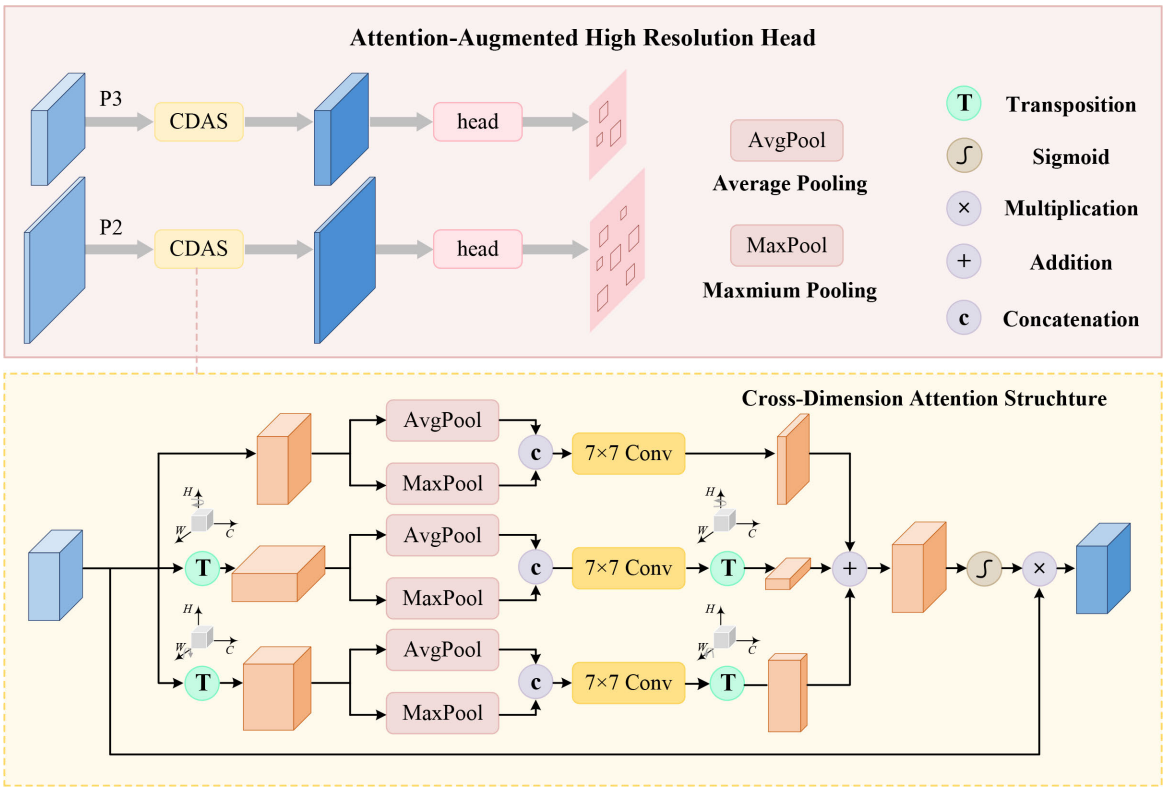


Fig. 3. Diagram of AMFFM. CDAS is introduced before prediction. Moreover, prediction layers are reconstructed with high-resolution P2 and P3 instead of P3, P4, and P5.

perceptual areas for neurons at different locations inside the network for the original input image. Based on the feature map $F_{\text{fusion}} \in \mathbb{R}^{C\times H\times W}$ obtained from AMFFM, we capture the cross-dimension dependencies through three branches. The top branch is the same as the spatial attention mechanism in CBAM [29]. The pixel dependencies along channel dimension are extracted by maximum and average pooling. Then the aggregated pooling results $F^{C}_{\text{pool}} \in \mathbb{R}^{2\times H\times W}$ is obtained by the concatenation operation. Followed by the 7×7 convolution to introduce the local spatial information, the saliency map along spatial dimension $S^{C} \in \mathbb{R}^{1\times H\times W}$ is generated. The generation process is formulated as follows:

$$S^{C} = \text{conv}_{7\times 7}[\text{MaxPool}(F_{\text{fusion}}), \text{AvePool}(F_{\text{fusion}})]. \tag{5}$$

The other branches capture the interdependencies along the length and width dimensions. The difference with the first branch is that we adjust the pooling direction by transposition operation, from which the interdependence between the channel and spatial dimensions can be captured. The transposition operations in the second and third branches change the index order from (C, H, W) to (H, C, W) and from (C, H, W) to (W, H, C), respectively. The elements in the transposed matrices F^{H}_{fusion} and F^{W}_{fusion} are written as

$$\begin{aligned} F^{H}_{\text{fusion}}(i, j, k) &= F_{\text{fusion}}(j, i, k) \\ F^{W}_{\text{fusion}}(i, j, k) &= F_{\text{fusion}}(k, j, i). \end{aligned} \tag{6}$$

The following steps are the same as that in the first branch. 7×7 convolution is also applied to generate the cross-dimension saliency maps S^{H} and S^{W}. Finally, we sum all the saliency maps using the broadcast mechanism and utilize the sigmoid activation function to obtain the pixel-by-pixel attention weights. The weights are assigned to the input feature maps for feature enhancement

$$F_{\text{enchaned}} = F_{\text{fusion}} \times \sigma\left(S^{H} + S^{W} + S^{C}\right). \tag{7}$$

For the YOLO-based architecture, the usual prediction layers for regression and classification are P3, P4, and P5, with downsampling levels of 8, 16, and 32. For an input image with a size of 640×640 pixels, the sizes of these layers are 80×80 pixels, 40×40 pixels, and 20×20 pixels. Taking a small target occupying 12×12 pixels for example, the corresponding areas in P3, P4, and P5 layers are 1.5×1.5 pixels, 0.75×0.75 pixels, and 0.375×0.375 pixels. The latter two are sub-pixel areas, which makes it difficult to achieve comprehensive detection in dense-oriented scenes. The visualization results of the feature maps extracted from different layers are shown in Fig. 4. P4 and P5 layers show less information representation of the target feature compared to the P2 and P3 layers, where the target regions corresponding to the original map are not highlighted.

Therefore, our method reconstructs the level of the prediction head with high resolution. We retain the P3 prediction layer and add the P2 prediction layer (with the downsampling levels of 4) to ensure comprehensive target detection in the densely distributed scenes. Besides, we remove the low-resolution P4 and P5 prediction layers to reduce the computation complexity.

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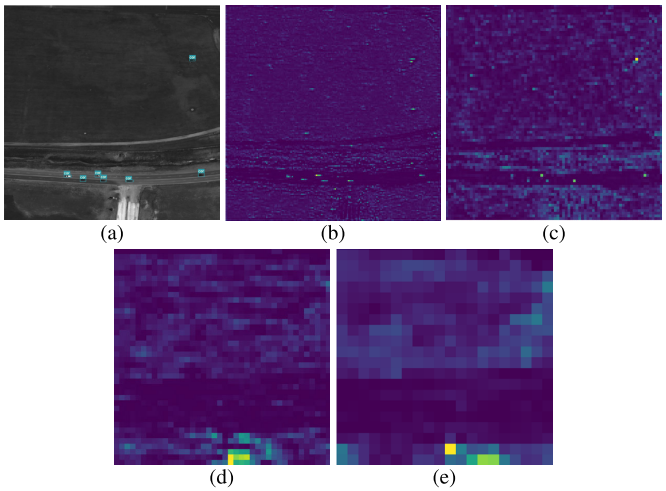


Fig. 4. Feature maps extracted from different layers. (a) Original image. Feature map from (b) P2 layer, (c) P3 layer, (d) P4 layer, and (e) P5 layer.

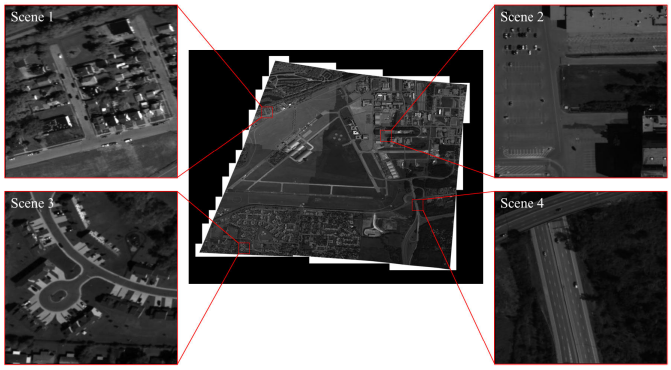


Fig. 5. Display of the scenes in UNICORN 2008.

IV. Experiments

A. Datasets

To comprehensively verify the performance of the proposed method, we choose two typical application scenarios, car detection and ship detection. We construct a small car dataset and select the public small ship dataset MASATI [38] and the VEDAI dataset [39] for subsequent experimental evaluation.

1) Small Car Dataset: The small car dataset is built based on the UNICORN 2008 dataset [40], which is composed of plenty of wide-area motion images. The original image size of the UNICORN dataset is around 10 000 × 10 000 pixels, while the average size of the car targets is only 14 × 14 pixels. We crop the images to 640 × 640 pixels and get a total of 3225 images. Then we label more than 30 000 car instances and randomly divide the images into training and test sets with the ratio of 7:3. The images cover various complex scenes, such as parking lots, roads, and communities, as shown in Fig. 5. There are a large number of suspected target areas in these scenes, which are prone to deteriorate the detection performance. Meanwhile, the targets with low contrast and limited appearance features are small in scale and weak in characteristics, which also increases the difficulty of detection. The target displays and size distributions are shown Fig. 6.

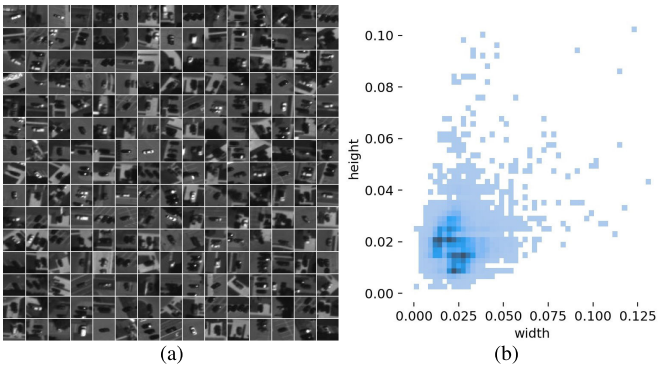


Fig. 6. Illustration of the targets in small car dataset. (a) Displays of small cars. (b) Size distributions with the average size of 14 × 14 pixels.

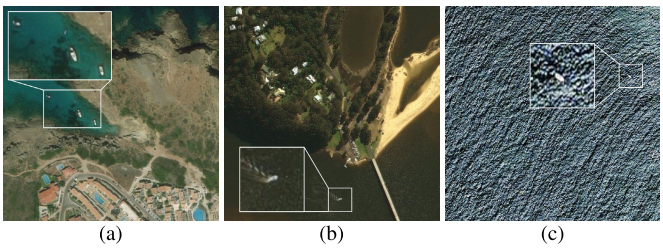


Fig. 7. Display of the scenes in small ship dataset. (a) Multiple targets. (b) Coast with false alarm sources similar to the targets. (c) Sea surface with strong cluttered interference.

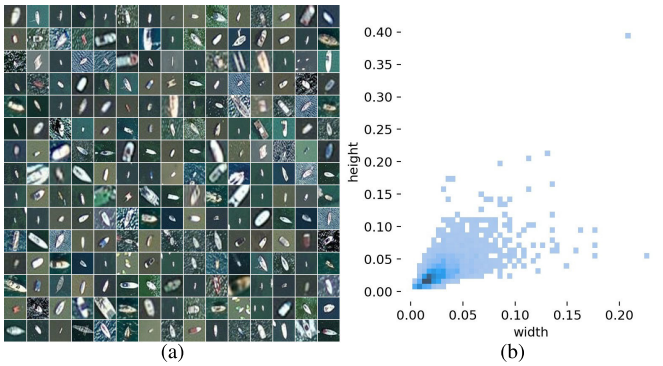


Fig. 8. Illustration of the targets in small ship dataset. (a) Displays of small ship. (b) Size distributions with the average size of 11 × 11 pixels.

2) Small Ship Dataset: The small ship detection experiment is conducted on the MASATI dataset. This dataset is composed of 7389 images with the size of 512 × 512 pixels, and the average size of the ship targets is around 11 × 11 pixels. We choose the images containing targets with 2368 images and around 7000 instances in total, then divide them into training sets and test sets with the ratio of 7:3. As shown in Fig. 7, the dataset contains various complex scenes with different imaging conditions like harbor, coast, and sea surface with strong cluttered interference. The target scale in the dataset is small and difficult to detect. Fig. 8 illustrates the target samples and the size distributions.

3) VEDAI Dataset: To further evaluate the detection performance, we also include a publicly available dataset called VEDAI and conduct comparison experiments to demonstrate the effectiveness of the proposed method. VEDAI comprises a diverse range of small vehicles, including cars, buses, and trucks. We only focus on detecting small targets, thus unifying

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TABLE I
RESULTS OF ABLATION EXPERIMENTS ON SMALL CAR DATASET

Methods	P	R	F1	AP	Params (M)	GFLOPs	FPS	TT (h)
YOLOv5x	84.55%	78.71%	0.8153	83.35%	86.17	204.0	18.6	17.2
YOLOv5l	84.00%	78.38%	0.8109	82.18%	46.11	107.8	31.3	8.9
YOLOv5m	82.58%	79.40%	0.8096	81.95%	20.85	47.9	49.1	5.8
YOLOv5s	82.86%	76.87%	0.7975	80.91%	7.01	15.8	66.3	4.2
YOLOv5n	77.33%	74.79%	0.7604	77.72%	1.76	4.2	77.4	2.8
YOLOv5n+AMFFM	83.24%	77.12%	0.8006	81.12%	2.28	5.8	72.6	3.4
YOLOv5n+AAHRH	82.69%	78.83%	0.8071	82.34%	1.24	4.2	71.9	3.5
YOLOv5n+AMFFM+AAHRH	**85.44%**	**80.90%**	**0.8310**	**85.73%**	**1.56**	**6.5**	**66.5**	**3.8**

all the categories as the car category. The dataset is split into training and test sets, with a ratio of 7:3 (865 images for training and 369 images for testing).

B. Experimental Settings

1) Evaluation Criteria: The detection performance is evaluated in terms of accuracy and speed. Precision (P), recall (R), $F1$-Score, and average precision (AP) are chosen to evaluate the detection accuracy. The computational complexity and inference speed are evaluated by the number of model parameters (Params), floating point operations (FLOPs), frames/s (FPS), and training time (TT).

Precision is defined as the percentage of correct predictions among all predictions, which can assess the degree of false alarms, while recall measures the degree of missed alarms. It is the percentage of correct predictions among the labeled targets

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{8}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{9}$$

where TP, FP, and FN represent the true positive, false positive, and false negative, respectively.

The $F1$ score is defined as the harmonic mean of the precision and recall. The AP is defined as the area enclosed by the curve of precision and recall. They are calculated as

$$F1 = \frac{2}{\frac{1}{P} + \frac{1}{R}} \tag{10}$$

$$\mathrm{AP} = \int_0^1 P(R) d(R). \tag{11}$$

Params and FLOPs are chosen to evaluate the computation complexity of models. The unit of Params is million, and the unit of FLOPs is GFLOPs (1 GFLOPs $= 10^9$ FLOPs). FPS can assess the real-time performance, and TT uses hour as the unit to evaluate the time cost for training.

2) Implementation Details: All the experiments are carried out on the workstation with an Inter Xeon[1] Silver 4210R CPU and an NVIDIA Quadro RTX 4000 GPU. We conduct our experiments with the Pytorch framework. For the comparison experiments based on YOLOv5, including YOLOv5n-x, TPH-YOLOv5, and the proposed method, we adopt the default parameters in the official codes except for the batch size. The batch size to train YOLOv5x and TPH-YOLOv5 is 4 instead of 8 due to the limitation of GPU memory. The input image size in the training process is 640 × 640 pixels for 300 epochs. We choose the stochastic gradient descent (SGD) optimizer. The initial learning rate is 0.01 with a weight decay of 0.0005 and momentum of 0.937. Mosaic data augmentation is applied to enrich the background, which splices four images into one for training. we adopt the default loss function from YOLOv5. The detection loss consists of three parts: localization loss, confidence loss, and classification loss, and $(\lambda_{\mathrm{loc}}, \lambda_{\mathrm{conf}}, \lambda_{\mathrm{cls}})$ is set to $(0.05, 1, 0.5)$ without additional modification

$$L_o = \lambda_{\mathrm{loc}} L_{\mathrm{loc}} + \lambda_{\mathrm{conf}} L_{\mathrm{conf}} + \lambda_{\mathrm{cls}} L_{\mathrm{cls}}. \tag{12}$$

The other comparison experiments are all implemented by MMDetection [41] with the default parameters, including the faster R-CNN, FCOS, YOLOF, etc. The randomly divided training set is used to train the network, while the test set is utilized for detection performance evaluation.

C. Ablation Analysis

To ensure high inference speed, we construct the network based on YOLOv5n with the lowest model complexity. We conduct the ablation analysis in small car detection and small ship detection to evaluate the effectiveness of the proposed method. Also, we simultaneously compare the detection performance of YOLOv5 with more computation to highlight our method's speed and accuracy advantages.

1) Small Car Detection: Table I illustrates the quantitative evaluation results of the ablation experiments for small car detection. Compared with the basic YOLOv5n network, the AMFFM and AAHRH can significantly improve the detection performance, which provide 3.40% and 4.62% increase in AP, respectively. Indeed, AMFFM increases the model parameters and reduces the inference speed by 4.8 FPS. Although the AAHRH does not cause the extra parameters, the addition of the P2 prediction layer achieves more fine-grained detection, thus reducing the inference speed by 5.5 FPS. The final AP gained by our method is 85.7%, which is increased by 8.1% compared to the baseline. As shown in Fig. 9, we visualize the AP-FPS curve and P-R curve to show our method's superior performance. The proposed method achieves a balance of accuracy and speed, which has a similar detection speed to YOLOv5s (66.5 versus 66.3 FPS) while outperforming the other YOLOv5 in accuracy, even the most complex YOLOv5x. Compared to YOLOv5l, the AP of the proposed method increases by 3.55%, and the FPS is 2.1× faster than that of

[1]Registered trademark.

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TABLE II
RESULTS OF ABLATION EXPERIMENTS ON SMALL SHIP DATASET

Methods	P	R	F1	AP	Params (M)	GFLOPs	FPS	TT (h)
YOLOv5x	86.02%	76.28%	0.8109	81.05%	86.17	204.0	19.2	11.3
YOLOv5l	85.71%	76.17%	0.8066	80.66%	46.11	107.8	31.7	5.7
YOLOv5m	84.60%	74.91%	0.7906	78.96%	20.85	47.9	50.3	3.9
YOLOv5s	81.33%	73.36%	0.7732	78.14%	7.01	15.8	66.9	2.7
YOLOv5n	78.25%	72.45%	0.7524	77.43%	1.76	4.2	78.4	1.8
YOLOv5n+AMFFM	85.58%	74.64%	0.7974	79.48%	2.28	5.8	73.1	2.2
YOLOv5n+AAHRH	84.12%	75.91%	0.7980	80.13%	1.24	4.2	72.7	2.3
YOLOv5n+AMFFM+AAHRH	**88.13%**	**77.01%**	**0.8220**	**82.84%**	**1.56**	**6.5**	**67.2**	**2.4**

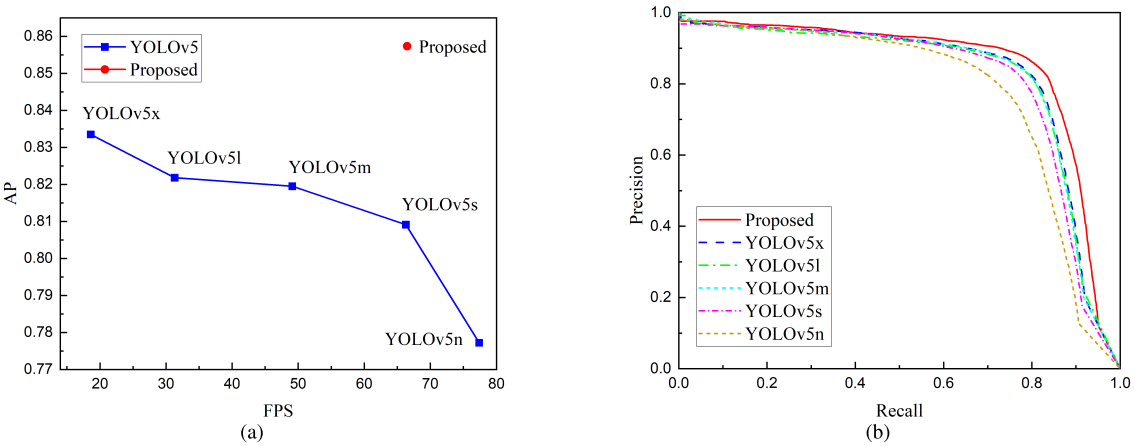


Fig. 9. Comparisons with YOLOv5 series for small car detection. (a) AP-FPS curve. (b) *P*-*R* curve.

YOLOv5l. Also, the Params and FLOPs are only 3.37% and 6.03% of the YOLOv5l's, respectively.

As depicted in Fig. 10, we visualize the detection results for some typical scenes to demonstrate the improvement intuitively. The baseline network cannot detect the targets occluded by the trees (the first row) or weak in the characteristics (the second row). In contrast, the proposed method can effectively improve the perception of small targets through AAHRH and detect all the targets. Meanwhile, the baseline network detects false alarm sources similar to the targets in geometric and grayscale features (the third and the fourth rows). In comparison, the proposed method can make full use of the semantic information through AMFFM to eliminate such false alarms.

2) Small Ship Detection: As shown in Table II, we provide the quantitative ablation experiment results of the small ship detection. The AP of the proposed method is 82.84%, which has a 5.41% improvement over the basic YOLOv5n. AMFFM can improve the AP by 2.05% and reduce the FPS by 5.3. Meanwhile, AAHRH increases AP by 2.70% with a 5.7 FPS decrease. Our method is advanced for all the accuracy metrics with the close inference speed of YOLOv5s (67.2 versus 66.9 FPS), as illustrated in Fig. 11. Taking YOLOv5l as a comparison, our method increases the AP by 2.18%, whose inference speed is 2.1× faster than that of YOLOv5l.

The detection results of the baseline and the proposed network are visualized in Fig. 12. The waves (the first row), reefs (the second row), and artificial facilities (the third row) in complex scenes interfere with the detection of the small ships because they show similar characteristics. The baseline network mistakes them as targets, which is avoided by our method. In the fourth row, the strong sea clusters weaken the characteristics of small ships. It affects the feature representation and results in missing alarms, which is solved by our method.

TABLE III
RESULTS OF ABLATION EXPERIMENTS WITH DIFFERENT ATTENTION MECHANISMS

Methods		P	R	F1	AP
Small Car Dataset	+SE	78.52%	75.89%	0.7718	78.19%
	+CBAM	78.64%	76.12%	0.7736	78.48%
	+CA	78.89%	76.77%	0.7782	79.62%
	+CDAS (ours)	**80.93%**	**77.13%**	**0.7898**	**80.71%**
Small Ship Dataset	+SE	81.39%	73.23%	0.7709	78.11%
	+CBAM	81.54%	73.69%	0.7742	78.36%
	+CA	81.79%	74.11%	0.7776	78.61%
	+CDAS (ours)	**82.81%**	**74.42%**	**0.7839**	**79.48%**

3) Comparisons With Different Attention Mechanisms: To fully illustrate the effectiveness of the proposed CDAS, we conduct several comparison experiments with the common attention mechanisms: SE [28], CBAM [29], and CA [32]. Table III shows that the proposed structure can achieve better performance than the others with the same condition. The other attention mechanisms ignore the cross-dimension interaction and contribute less to distinguishing small targets from complicated backgrounds. Compared to CA, the proposed CDAS improves the AP with additional 1.09% and 0.87% increases on the small car and ship dataset.

D. Algorithm Performance Comparison

To verify the effectiveness and robustness of the proposed method, we compare our approach with several state-of-art

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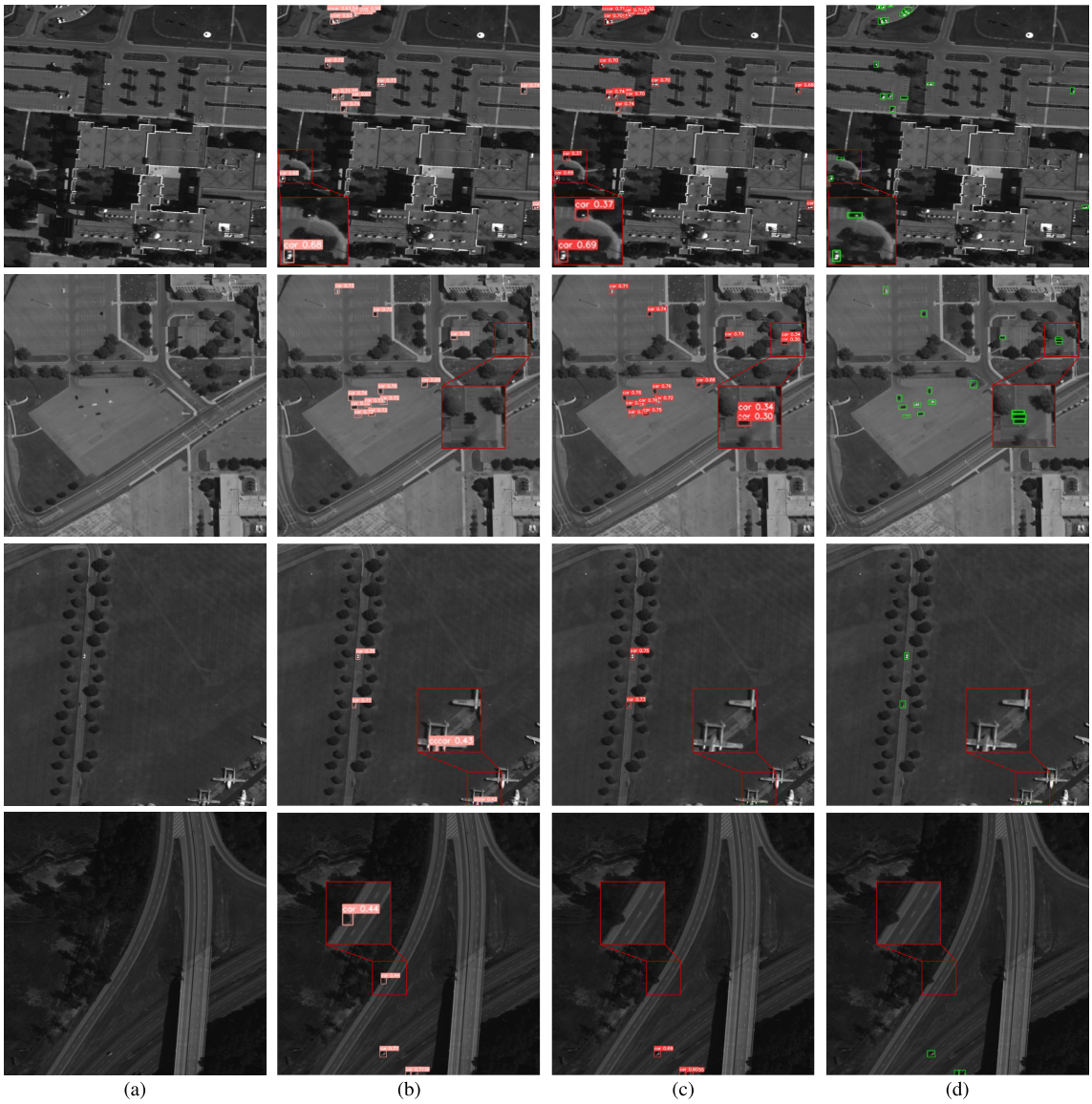


Fig. 10. Visualization of the dilated experiments for small car detection. (a) Input image. (b) YOLOv5. (c) Our method. (d) Ground truth.

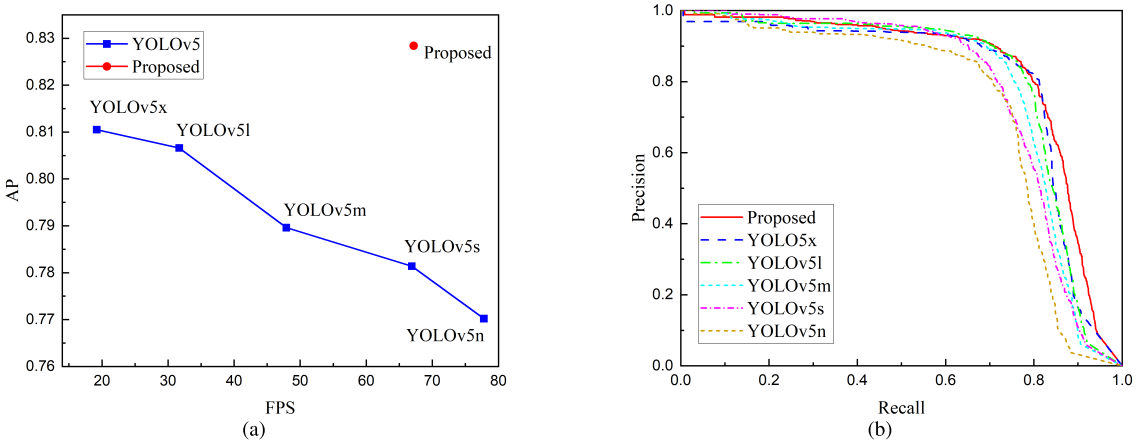


Fig. 11. Comparisons with YOLOv5 series for small ship detection. (a) AP-FPS curve. (b) *P*-*R* curve.

detectors, including some one-stage detectors, two-stage detectors, and anchor-free detectors. We evaluate the detection performance in the application scenarios of small car and small ship detection.

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TABLE IV
COMPARISON RESULTS WITH TYPICAL DETECTORS ON SMALL CAR DATASET

Methods		P	R	F1	AP	FPS	TT (h)
Two-stage detectors	Faster R-CNN [42]	77.87%	68.05%	0.7263	72.91%	9.1	5.2
	Cascade R-CNN [43]	80.98%	69.11%	0.7458	75.60%	7.7	6.9
	Grid R-CNN [44]	80.47%	66.96%	0.7310	73.58%	8.4	9.0
	RoI Transformer [45]	81.53%	70.36%	0.7553	77.72%	6.6	11.2
Anchor-free detectors	FCOS [46]	76.12%	73.79%	0.7493	75.49%	10.7	14.3
	CentripetalNet [47]	74.59%	77.12%	0.7583	76.19%	9.7	26.3
	YOLOX [48]	80.91%	71.86%	0.7612	78.73%	21.4	19.2
	Oriented RepPoints [49]	82.66%	78.86%	0.8072	83.42%	9.6	23.3
One-stage detectors	YOLOv3 [21]	77.13%	78.05%	0.7759	73.53%	17.0	15.8
	RetinaNet [19]	66.26%	70.43%	0.6828	62.01%	9.3	28.9
	YOLOF [50]	53.25%	39.84%	0.4558	37.40%	13.8	16.4
	S2A-Net [51]	79.23%	79.88%	0.7955	76.43%	8.6	30.0
	EfficientDet [52]	76.91%	78.79%	0.7778	75.42%	10.4	33.8
	TPH-YOLOv5 [53]	80.86%	74.96%	0.7780	80.69%	24.4	15.1
	Proposed method	**85.44%**	**80.90%**	**0.8310**	**85.73%**	**66.5**	**3.8**



(a) (b) (c) (d)

Fig. 12. Visualization of the dilated experiments for small ship detection. (a) Input image. (b) YOLOv5. (c) Our method. (d) Ground truth.

1) Small Car Detection: As shown in Table IV and Fig. 13, the proposed method performs optimally both in the inference speed and detection accuracy. Among those methods, the two-stage detectors with coarse-to-fine pipelines do not show

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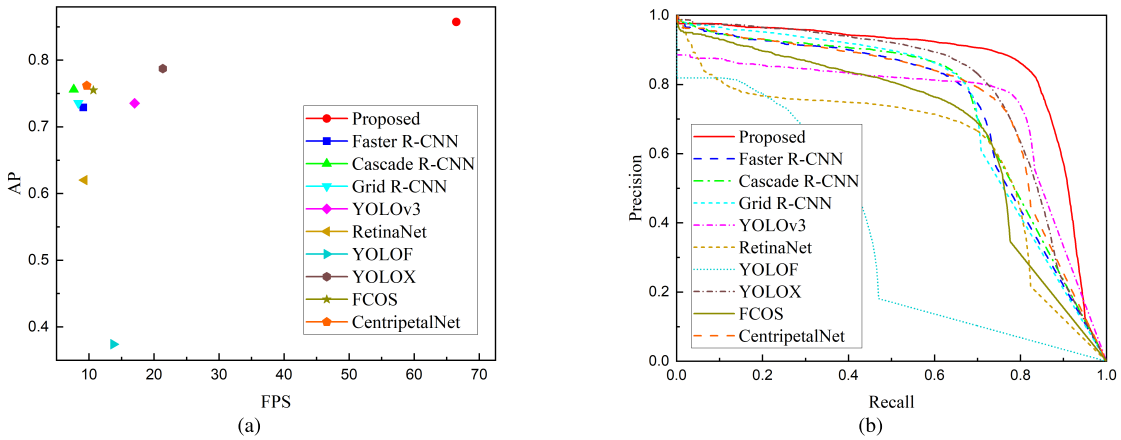


Fig. 13. Comparisons with typical networks for small car detection. (a) AP-FPS curve. (b) *P*-*R* curve.

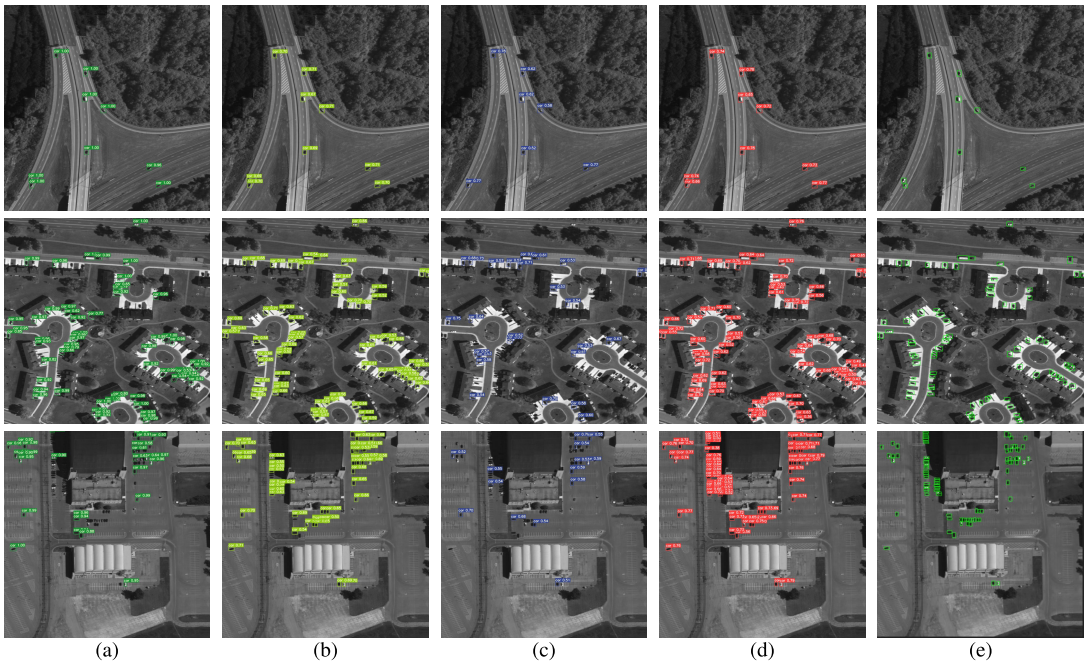


Fig. 14. Examples of detection results obtained by our method and the typical networks for small car detection. (a) Cascade-R-CNN. (b) YOLOv3. (c) FCOS. (d) Our method. (e) Ground truth.

superiority for small car detection, and the one-stage detectors have an overall speed advantage. Compared to Oriented RepPoints with the best performance among the other typical networks, the proposed method has a 2.31% increase in AP with a 6.9× higher inference speed.

We choose the two-stage network Cascade R-CNN, the one-stage network YOLOv3, and the anchor-free network FCOS to visualize the detection results and compare them with the proposed network. The road, complex community, and densely distributed scenes are selected to display the detection performance comprehensively, as illustrated in Fig. 14. It can be concluded that FCOS does not rely on predefined anchors but has many missing alarms. Based on the region proposal network and the cascade detection, Cascade R-CNN obtains higher confidence for target detection with slight interference from the complex scenes but has more missing alarms than YOLOv3. The proposed network enhances the perception ability of small targets by AAHRH and performs well for dense-oriented target detection. Besides, combined with AMFFM, our method is not prone to be interfered with by false alarm sources similar to the targets and obtains optimal detection performance.

2) Small Ship Detection: For small ship detection, the proposed method can ensure optimal detection accuracy and the fastest inference speed. As shown in Table V and Fig. 15, RoI Transformer has the highest detection accuracy among the typical networks. Compared to that, the AP of our method is 82.73% increased by 3.01%, and the inference speed is 10.3× faster than that of RoI Transformer.

It can be seen in Fig. 16 that the two-stage network Cascade R-CNN performs better for small ship detection compared with the one-stage network YOLOv3 and the anchor-free network FCOS. It can achieve higher recall for targets interfered by the sea wave or with weak characteristics but cannot

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TABLE V
COMPARISON RESULTS WITH TYPICAL DETECTORS ON SMALL SHIP DATASET

Methods		P	R	F1	AP	FPS	TT(h)
Two-stage detectors	Faster R-CNN [42]	82.14%	70.25%	0.7573	77.50%	9.0	4.0
	Cascade R-CNN [43]	84.63%	72.56%	0.7813	78.80%	7.6	5.1
	Grid R-CNN [44]	82.75%	71.23%	0.7656	76.47%	8.8	6.8
	RoI Transformer [45]	85.23%	73.36%	0.7885	79.83%	6.5	9.1
Anchor-free detectors	FCOS [46]	77.96%	73.63%	0.7573	73.44%	10.7	10.6
	CentripetalNet [47]	75.23%	76.51%	0.7586	76.23%	9.5	19.5
	YOLOX [48]	83.78%	72.56%	0.7777	78.76%	20.5	14.8
	Oriented RepPoints [49]	83.98%	73.32%	0.7829	78.77%	9.6	16.8
One-stage detectors	YOLOv3 [21]	74.76%	72.65%	0.7369	69.41%	16.7	12.0
	RetinaNet [19]	78.87%	73.26%	0.7596	72.65%	9.2	21.6
	YOLOF [50]	68.36%	57.62%	0.6253	50.18%	13.7	12.2
	S2A-Net [51]	79.81%	74.59%	0.7711	77.74%	8.4	23.8
	EfficientDet [52]	79.23%	74.48%	0.7678	76.47%	10.3	26.5
	TPH-YOLOv5 [53]	84.54%	73.85%	0.7883	79.39%	24.1	10.0
	Proposed method	**88.13%**	**77.01%**	**0.8220**	**82.84%**	**67.2**	**2.4**

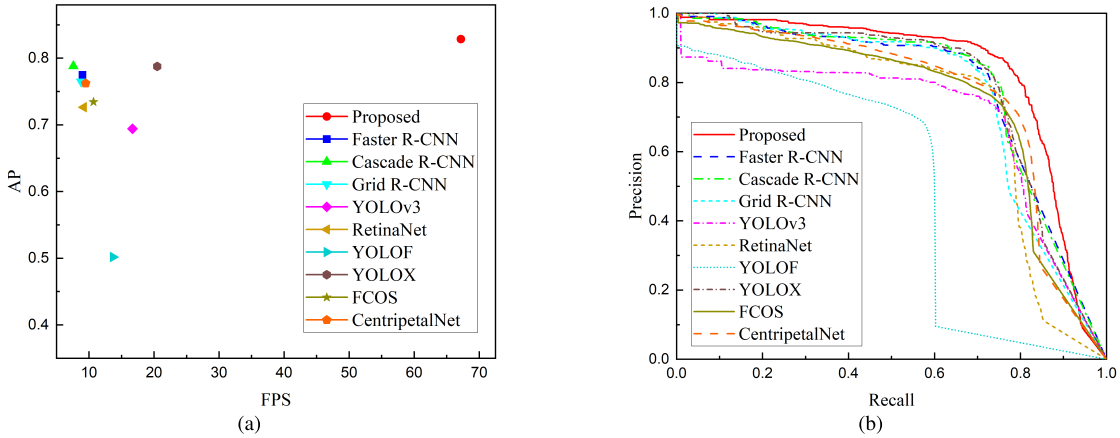


Fig. 15. Comparisons with typical networks for small ship detection. (a) AP-FPS curve. (b) P-R curve.

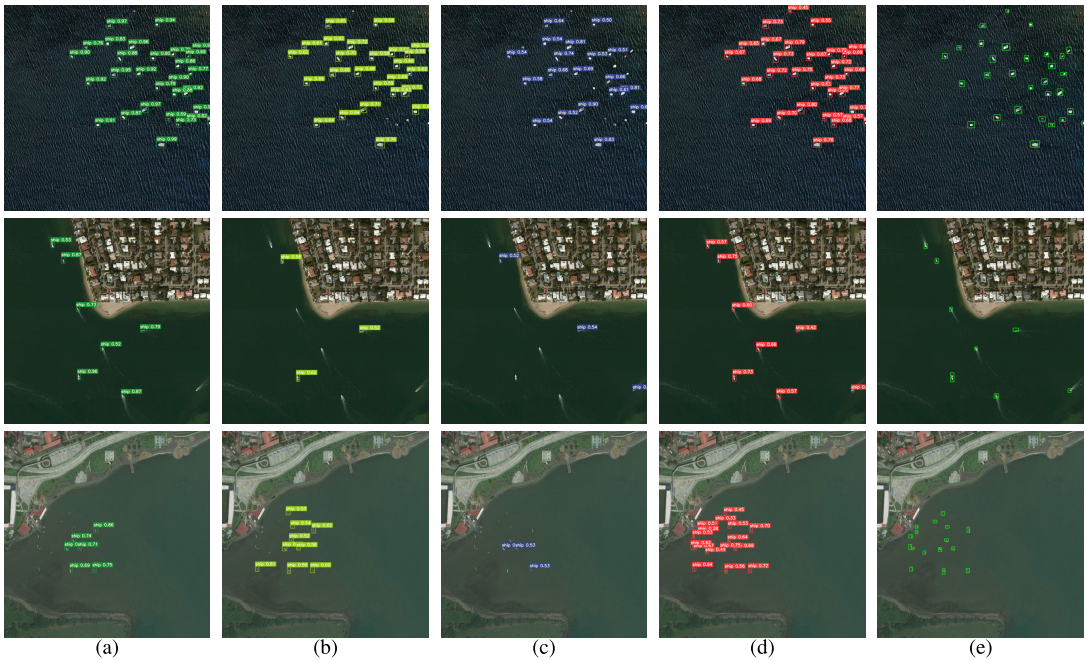
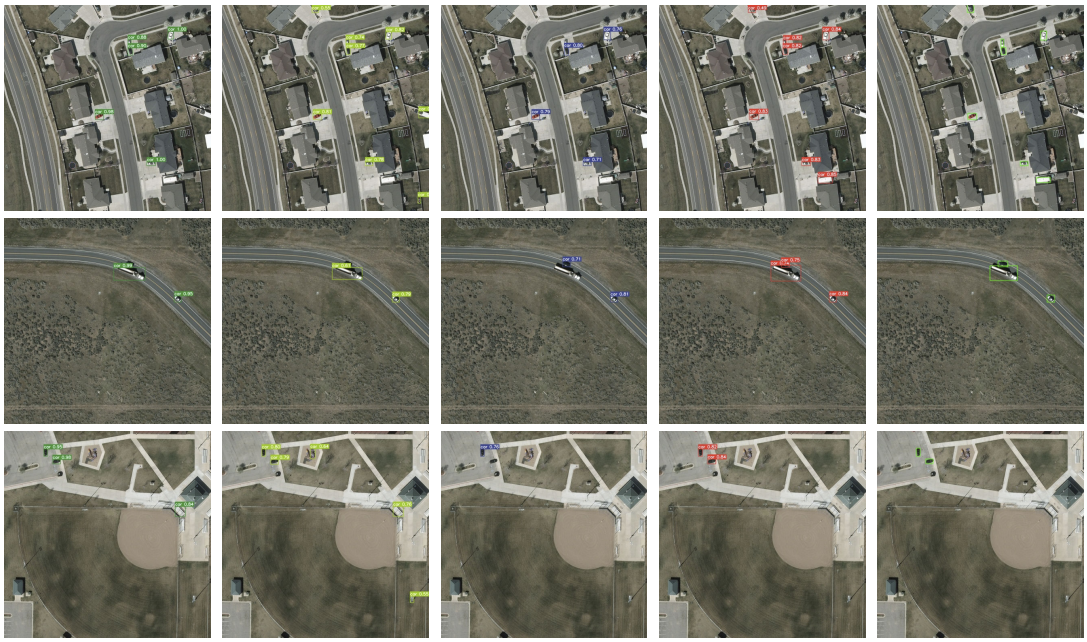


Fig. 16. Examples of detection results obtained by our method and the typical networks for small ship detection. (a) Cascade-R-CNN. (b) YOLOv3. (c) FCOS. (d) Our method. (e) Ground truth.

guarantee the detection in densely distributed scenes. The proposed network adopts AMFFM and AAHRH, which has a good perception ability for small targets and achieve promising detection for dense-oriented scenes.

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(a) (b) (c) (d) (e)

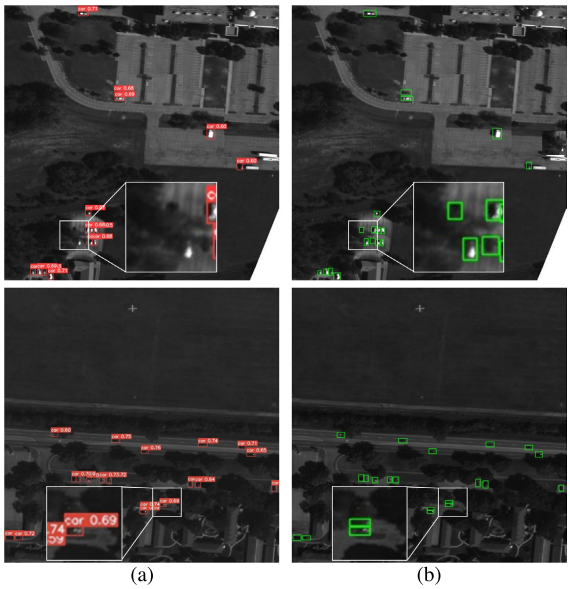
Fig. 17. Examples of detection results obtained by our method and the typical networks on VEDAI dataset. (a) Cascade-R-CNN. (b) YOLOv3. (c) FCOS. (d) Our method. (e) Ground truth.

TABLE VI
COMPARISON RESULTS WITH TYPICAL DETECTORS ON VEDAI DATASET

Methods	P	R	F1	AP	FPS
Faster R-CNN [42]	85.45%	78.91%	0.8205	83.52%	8.9
Cascade R-CNN [43]	88.29%	82.79%	0.8545	86.67%	7.7
Grid R-CNN [44]	86.94%	80.11%	0.8339	83.29%	8.6
FCOS [46]	81.21%	82.34%	0.8177	80.64%	10.6
CentripetalNet [47]	79.02%	84.52%	0.8168	82.45%	9.5
YOLOX [48]	86.68%	81.26%	0.8388	85.14%	20.8
YOLOv3 [21]	77.82%	79.41%	0.7861	76.15%	16.8
RetinaNet [19]	82.87%	82.37%	0.8262	79.81%	9.1
YOLOF [50]	71.11%	64.91%	0.6787	57.69%	13.8
Proposed method	**91.30%**	**87.47%**	**0.8934**	**90.24%**	**67.0**

3) Extension Comparsion Experiments on VEDAI Dataset: As shown in Table VI, the proposed method performs optimally in the detection accuracy and inference speed. Compared to Cascade R-CNN, with the best performance among the other typical networks, the proposed method has a 3.57% increase in AP with an 8.7× higher inference speed. Fig. 17 illustrates the detection results of the proposed and typical methods. It is evident that our method can not only alleviate the inference of alarm sources with similar characteristics (third row in Fig. 17), but also achieve high recall in complex scenes (first row in Fig. 17) and densely distributed scenes (second row in Fig. 17).

The main limitation is that our proposed method may not be suitable for detecting heavily occluded targets. For example, small targets obscured by trees, buildings, and shadows over half of the area are challenging to detect due to the limited features. Fig. 18 illustrates the missing alarms caused by the occlusion of the trees. The heavily occluded small target detection could be a potential area for future research. By exploring effective contextual information, this problem could be alleviated to further improve the effectiveness and applicability of our proposed method.



(a) (b)

Fig. 18. Missing alarms for the heavily occluded targets. (a) Proposed method. (b) Ground truth.

V. CONCLUSION

In this article, we propose a novel detection framework to detect small targets in complicated remote sensing images. An AMFFM and an AAHRH are proposed to alleviate the challenges of small target detection. Specifically, AMFFM is put forward to suppress the interference of false alarm sources in complicated scenes. First, we enhance the multi-level features where the high-level features come from the

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context modeling for semantic information, and the low-level features are refined by the reference map with little noise interference. Then the enhanced multilevel features are fused based on the spatial and channel significance adaptively. Afterward, AAHRH is designed to enhance the perception of small targets by embedding cross-dimension interaction with the attention mechanism. We reconstruct the prediction heads with high-resolution layers to improve the detection performance for densely distributed scenes. The experimental results conducted on the constructed small car dataset, the public small ship dataset, and the VEDAI dataset illustrate that the proposed method realizes the state-of-the-art performance, which outperforms in all the evaluation metrics. However, the main limitation is that our proposed method may not be suitable for detecting heavily occluded targets, which could be a potential research area. For example, small targets obscured by trees, buildings, and shadows over half of the area are challenging to detect due to the limited features. In the future work, we plan to explore effective contextual information further to improve the effectiveness and applicability of our proposed method and conduct research on fine-grained small target recognition.

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